

Solved Model Paper ■ Managerial Economics

PART A — MCQs

1. What is the primary goal of managerial economics?

- A) Maximize profits B) Minimize costs
- C) Maximize sales D) Maximize shareholder value

Answer: A) Maximize profits

Explanation:

Managerial economics focuses on optimal decision-making to achieve **profit maximization**, though modern firms may also consider growth and shareholder value.

2. Which of the following best describes managerial economics?

- A) Study of economic policies
- B) Application of economic theories and tools of decision analysis to business decision-making
- C) Analysis of government regulations
- D) Study of consumer behavior

Answer: B) Application of economic theories and tools of decision analysis to business decision-making

Explanation:

Managerial economics applies microeconomic theory and quantitative tools to business decisions.

3. What is the main focus of microeconomics in managerial economics?

- A) Aggregate economy B) Individual firm or market
- C) International trade D) Economic growth

Answer: B) Individual firm or market

Explanation:

Microeconomics studies firm behavior, pricing, demand, and supply.

4. Why is decision-making important in managerial economics?

- A) To minimize risks B) To maximize profits
- C) To allocate resources efficiently D) All of the above

Answer: D) All of the above

Explanation:

Decision making helps:

1. Minimize risks

2. Maximize profits
3. Allocate resources efficiently

5. Opportunity cost refers to:

- A) The cost of producing good
- B) The value of the next best alternative forgone
- C) The cost of raw materials
- D) The cost of labor

Answer: B) The value of the next best alternative forgone

Explanation:

Opportunity cost is what you sacrifice when choosing one option.

6. Which branch of economics is most closely related to managerial economics?

- A) Macroeconomics
- B) Microeconomics
- C) International Economics
- D) Development Economics

Answer: B) Microeconomics

7. What is the primary goal of optimization?

- A) Minimize cost
- B) Maximize profits
- C) Maximize revenue
- D) Maximize market share

Answer: B) Maximize profits

8. Profit of the firm is maximum at:

- A) $MR = MC$
- B) $MR > MC$
- C) $MR < MC$
- D) none of the above

Answer: A) $MR = MC$

Explanation:

Profit maximization occurs when **Marginal Revenue = Marginal Cost**

9. What do we mean by Marginal revenue?

- A) Change in total cost from producing one more unit
- B) Change in total revenue from selling one more unit
- C) Total revenue divided by quantity sold
- D) Average revenue per unit

Answer: B) Change in total revenue from selling one more unit

10. At optimal level of output total revenue curve is:
A) Minimum B) Increasing C) Decreasing D) Maximum

Answer: D) Maximum

11. If $MR > MC$ which strategy firm will adopt?
A) Minimum B) Increasing C) Decreasing D) Maximum

Answer: A) Increase production

Explanation:

If marginal revenue exceeds marginal cost, producing more increases profit.

12. The point where $MC = MR$ is called:
A) Break-even point B) Shut-down point C) Profit-maximizing point D) Equilibrium point

Answer: C) Profit-maximizing point

13. If $MR = 90$ and $MC = 70$, firm should:
A) Increase output B) Decrease output C) Keep output same D) Shut down

Answer: A) Increase output

14. Optimization techniques help in:
A) Pricing decisions B) Production decisions C) Resource allocation D) All of the above

Answer: D) All of the above

15. Break-even point represents:
A) Maximum profit B) Minimum cost C) Where $TR = TC$ D) Where $MR = MC$

Answer: C) Where $TR = TC$

16. At break-even point:
A) Firm makes profit B) Firm incurs loss C) Firm covers all costs D) Firm shuts down

Answer: C) Firm covers all costs

17. If sales below break-even:
A) Firm makes profit B) Firm incurs loss C) Firm breaks even D) Firm maximizes profit

Answer: B) Firm incurs loss

18. Break-even helps firms decide:
A) Production levels B) Pricing strategy C) Sales targets D) all A, B and C

Answer: D) All A, B and C

19. Price elasticity > 1 means:

- A) elastic B) inelastic C) Demand is unitary elastic D) unrelated

Answer: A) Elastic

20. Which good has inelastic demand?

- A) Luxury cars B) Salt C) Air travel D) Designer clothes

Answer: B) Salt

21. If demand falls by 20% due to increase in price by 10% demand is:

- A) Inelastic B) Unit elastic C) Perfectly elastic D) Elastic

$PED = \% \text{ change in demand} / \% \text{ change in price}$

$$PED = 20 / 10 = 2$$

Answer: D) Elastic

22. Cross price elasticity measures:

- A) Response of demand to income change
B) Response of demand to price change of another good
C) Response of demand to price change of the same good
D) None of the above

Answer: B) Response of demand to price change of another good

23. If income rises 10% and demand increases 20%, income elasticity is:

- A) 0.5 B) 1 C) 2 D) -2

$$\text{Income elasticity} = 20 / 10 = 2$$

Answer: C) 2

24. If Price elasticity of Demand is 0.5, this means demand is:

- A) Elastic B) Inelastic C) Unit elastic D) Perfectly elastic

Answer: B) Inelastic

25. Which factor increases demand elasticity?

- A) Necessity good B) Many substitutes C) Low price D) Short run

Answer: B) Many substitutes

26. If two goods are complements, cross elasticity is:

- A) Positive B) Negative C) Zero D) Infinite

Answer: B) Negative

27. For a luxury good, income elasticity is:

- A) < 0 B) $0 < IE < 1$ C) > 1 D) $= 1$

Answer: C) > 1

28. If demand is perfectly elastic, PED is:

- A) 0 B) 1 C) Infinite D) Negative

Answer: C) Infinite

29. A movement along the demand curve is caused by:

- A. A change in consumer income.
B. A change in tastes and preferences.
C. A change in the price of the good itself.
D. A change in the price of related goods.

Answer: C) Change in price of the good itself

30. If an increase in consumer income leads to a decrease in the demand for good X, then good X is:

- A. A normal good. B. A substitute good.
C. A complementary good. D. An inferior good.

Answer: D) Inferior good

PART B — Questions

Q1 (a) Define Managerial Economics. Briefly explain the relationship of managerial economics with other disciplines.

- **Definition of Managerial Economics**

Managerial Economics is the application of economic theory and analytical tools to solve business problems and assist managers in decision-making.

It helps managers in:

- Pricing decisions
- Cost analysis
- Production planning
- Demand forecasting
- Profit maximization

Example

A manager deciding whether to increase production uses:

- Demand analysis
- Cost analysis
- Profit forecasting

This is managerial economics in practice.

- **Nature of Managerial Economics**

Managerial economics is:

- Practical in nature
- Decision-oriented
- Microeconomic in approach
- Interdisciplinary

- **Relationship with Other Disciplines**

1. Economics

Managerial economics is based on economic theory.

Microeconomics

Helps in:

- Demand analysis
- Pricing decisions
- Cost analysis
- Market structure

Macroeconomics

Helps in:

- Inflation analysis
- Business cycle
- National income
- Government policy

2. Mathematics

Mathematics helps in:

- Optimization
- Profit maximization
- Cost functions
- Demand functions

Example:

$$\text{Profit} = \text{TR} - \text{TC}$$

3. Statistics

Statistics helps in:

- Forecasting demand
- Market analysis
- Regression analysis

- Sampling

4. Accounting

Accounting provides:

- Cost data
- Profit data
- Financial statements

Managers use this data for decision making.

5. Management Theory

Management theory helps in:

- Planning
- Organizing
- Decision-making
- Control

Q1 (b) Explain briefly Theory of the Firm

- **Theory of the Firm**

The theory of the firm explains how firms make decisions regarding:

- Output
- Pricing
- Cost
- Profit

- **Objectives of Firm**

1. Profit Maximization

Traditional objective:

Firm aims to maximize profit.

Profit = TR – TC

2. Sales Maximization

Some firms aim to maximize sales instead of profit.

3. Growth Maximization

Large firms aim to increase:

- Market share
- Size
- Revenue

4. Market Share Maximization

Firms aim to dominate market.

Q2 (a) Define Profit. Differentiate between Business Profit and Economic Profit. Briefly elaborate different theories of profit.

- **Definition of Profit**

Profit is the difference between total revenue and total cost.

$\text{Profit} = \text{Total Revenue} - \text{Total Cost}$

- **Types of Profit**

1. Business Profit (Accounting Profit)

$\text{Business Profit} = \text{Total Revenue} - \text{Explicit Cost}$

Explicit costs include:

- Wages
- Rent
- Raw materials
- Utilities

Example

Revenue = 100,000

Explicit cost = 70,000

Profit = 30,000

2. Economic Profit

$\text{Economic Profit} = \text{Total Revenue} - (\text{Explicit} + \text{Implicit Cost})$

Implicit costs include:

- Owner's salary
- Opportunity cost
- Capital cost

Example

Revenue = 100,000

Explicit cost = 70,000

Implicit cost = 20,000

Economic Profit = 10,000

- **Theories of Profit**

1. Risk Bearing Theory (Hawley)

Profit is reward for taking risk.

Business risks include:

- Market risk
- Price risk
- Competition

2. Innovation Theory (Schumpeter)

Profit arises due to innovation.

Examples:

- New products
- New technology
- New markets

3. Uncertainty Bearing Theory (Knight)

Profit is reward for uncertainty.

Uncertainty includes:

- Demand changes
- Technology changes
- Government policy

4. Monopoly Profit Theory

Firms earn profit due to:

- Market power
 - Lack of competition
-

Q2 (b) Write a short note on Business Ethics

- **Business Ethics**

Business ethics refers to moral principles guiding business activities.

- **Importance of Business Ethics**

1. Fair Pricing

Avoid charging unfair prices.

2. Honest Advertising

Avoid misleading advertisements.

3. Consumer Protection

Provide quality goods.

4. Employee Welfare

Provide fair wages.

5. Environmental Protection

Avoid pollution.

Q3 (a) Explain the Law of Demand and its Exceptions

- **Law of Demand**

The law of demand states:

"Other things remaining constant, when price increases demand decreases, and when price decreases demand increases."

- **Assumptions**
 - Income constant
 - Taste constant
 - Price of related goods constant

- **Demand Schedule**

Price	Demand
10	100

20	80
30	60

- **Demand Curve**

Downward sloping curve.

- **Exceptions to Law of Demand**

1. Giffen Goods

Demand increases when price increases.

Example: Inferior food items

2. Prestige Goods

Luxury goods

Example:

- Gold
- Diamonds

3. Expectations

Future price rise increases demand.

4. Ignorance

Consumers assume higher price means better quality.

Q3 (b) Determinants of Demand

- **Major Determinants of Demand**

1. Price of Product

Price ↑ Demand ↓

2. Income

Income ↑ Demand ↑

3. Taste and Preference

Fashion changes demand.

4. Population

Population ↑ Demand ↑

5. Substitute Goods

Tea and coffee

Price of tea ↑ demand for coffee ↑

6. Complementary Goods

Car and petrol

Price of car ↑ petrol demand ↓ .

Q4 Explain Elasticity of Demand

- **Elasticity of Demand**

Elasticity of demand measures responsiveness of demand to changes in price, income, or related goods.

- **Types of Elasticity**

1. Price Elasticity of Demand

$PED = \% \text{ Change in Quantity} / \% \text{ Change in Price}$

Types

Elastic (>1)

Inelastic (<1)

Unitary ($=1$)

2. Income Elasticity

Income elasticity measures demand change due to income change.

Income elasticity = $\% \Delta Q / \% \Delta Y$

3. Cross Elasticity

Measures demand change due to price of related goods.

Cross elasticity = $\% \Delta Q / \% \Delta P$ of related goods

Q5 Marginal Analysis and Optimal Decision Making

Marginal Analysis

Marginal analysis compares:

- Marginal Revenue (MR)
- Marginal Cost (MC)

Decision Rule

Profit maximization when:

$MR = MC$

Relationship Between TR and MR

MR = Change in Total Revenue

If $MR > 0 \rightarrow$ TR increasing

Relationship Between TC and MC

MC = Change in Total Cost

Example

MR = 50

MC = 30 $\rightarrow$ Increase output

$$MR = 30$$

$$MC = 50 \rightarrow \text{Reduce output}$$

Q6 How the firm maximizes profit? Using a diagram to illustrate the optimal level of output where $MR=MC$.

Firm maximizes profit when:

$$MR = MC$$

Conditions

1. $MR = MC$
2. MC cuts MR from below

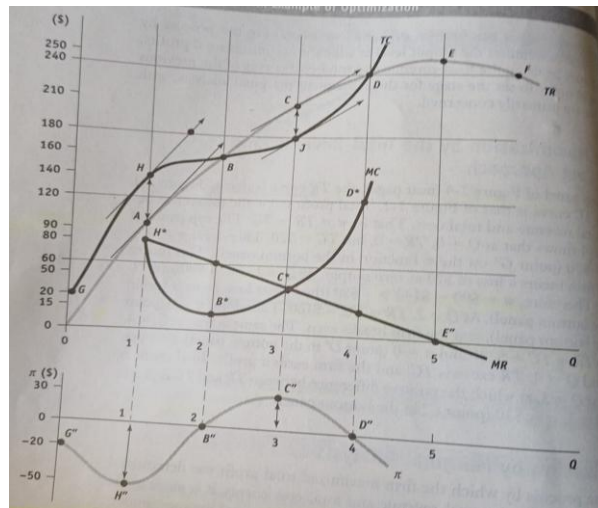
Example

$$MR = 40$$

$$MC = 40$$

Profit maximized.

Q7 Explain the diagram mentioned below:



Identify the break-even points. Illustrate the area between the breakeven points and the profit maximizing level of output.

Answer:

Break-even occurs when:

$$\text{Total Revenue} = \text{Total Cost}$$

Two Break Even Points

1. First break-even point
2. Second break-even point

Profit Area

Between break-even points = Profit

Loss Area

Outside break-even points = Loss

Q8 Find out the profit maximizing level of output of the following function:

$$TR = 4000Q - 33Q^2 \quad TC = 2Q^3 - 3Q^2 + 400Q + 5000$$

Given

$$TR = 4000Q - 33Q^2$$

$$TC = 2Q^3 - 3Q^2 + 400Q + 5000$$

Step 1 Find MR

$$MR = d(TR)/dQ$$

$$MR = 4000 - 66Q$$

Step 2 Find MC

$$MC = 6Q^2 - 6Q + 400$$

Step 3 MR = MC

$$4000 - 66Q = 6Q^2 - 6Q + 400$$

Solve

$$Q = 20$$

Profit Maximizing Output = 20 units.

Q9 What is differentiation? Explain the different rules of derivatives with the help of examples.

Introduction to Differentiation

Differentiation is a mathematical technique used to measure rate of change of one variable with respect to another variable. In managerial economics, differentiation is mainly used for:

- Profit maximization
- Cost minimization
- Revenue maximization
- Optimization problems

For example:

- Marginal Cost = derivative of Total Cost
- Marginal Revenue = derivative of Total Revenue
- Profit maximization occurs when **MR = MC**

Thus, differentiation plays an important role in decision-making in managerial economics.

Basic Concept of Differentiation

If:

$$Y = f(x)$$

Then derivative is:

$$\frac{d}{dx} (x^n) = nx^{n-1}$$

This means:

- dy = small change in y
- dx = small change in x
- dy/dx = rate of change

Rules of Differentiation

There are five major rules of differentiation:

1. Power Rule
2. Constant Rule
3. Sum Rule
4. Product Rule
5. Quotient Rule

1. Power Rule

Formula

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

This rule states that:

- Bring the power down
- Subtract 1 from the power

Examples

Example 1

If:

$$Y = x^2$$

Then:

$$dY/dx = 2x$$

Example 2

If:

$$Y = x^3$$

Then:

$$dY/dx = 3x^2$$

Example 3

If:

$$Y = 5x^4$$

Then:

$$dY/dx = 20x^3$$

Explanation:

- Multiply coefficient by power
- Reduce power by 1

2. Constant Rule

Formula

$$\frac{d}{dx} (c) = 0$$

This means derivative of a constant is always zero.

Examples

Example 1

$$Y = 10$$

Derivative = 0

Example 2

$$Y = 500$$

Derivative = 0

Reason:

Constant values do not change, so their rate of change is zero.

3. Sum Rule

If function contains addition or subtraction:

Formula

$$\frac{d}{dx} [f(x) + g(x)] = f'(x) + g'(x)$$

Differentiate each term separately.

Example 1

$$Y = x^2 + x$$

Derivative:

$$= 2x + 1$$

Example 2

$$Y = x^3 + 5x^2 + 10$$

Derivative:

$$= 3x^2 + 10x + 0$$

$$= 3x^2 + 10x$$

4. Product Rule

When two functions are multiplied:

Formula

$$\frac{d}{dx} (uv) = u'v + uv'$$

Steps:

1. Differentiate first function
2. Multiply by second function
3. Add first function multiplied by derivative of second

Example

$$Y = x^2 (x + 1)$$

$$\begin{aligned}
 &\text{Let} \\
 &u = x^2 \\
 &v = x + 1 \\
 &\text{Then:} \\
 &u' = 2x \\
 &v' = 1 \\
 &\text{Now apply formula:} \\
 &= (2x)(x+1) + (x^2)(1) \\
 &= 2x^2 + 2x + x^2 \\
 &= 3x^2 + 2x
 \end{aligned}$$

5. Quotient Rule

Used when one function is divided by another.

Formula

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{vu' - uv'}{v^2}$$

Steps:

1. Denominator $\times$ derivative of numerator
2. Minus numerator $\times$ derivative of denominator
3. Divide by denominator squared

Example

$$\begin{aligned}
 Y &= x^2 / x \\
 &\text{Let} \\
 &u = x^2 \\
 &v = x \\
 &\text{Then} \\
 &u' = 2x \\
 &v' = 1 \\
 &\text{Apply formula:} \\
 &= (x)(2x) - (x^2)(1) / x^2 \\
 &= 2x^2 - x^2 / x^2 \\
 &= x^2 / x^2 \\
 &= 1
 \end{aligned}$$

Application of Differentiation in Managerial Economics

Differentiation is used to find:

Marginal Cost

If:

$$TC = 5Q^2 + 10Q + 100$$

Then:

$$MC = d(TC)/dQ$$

$$MC = 10Q + 10$$

Marginal Revenue

If:

$$TR = 100Q - 2Q^2$$

Then:

$$MR = 100 - 4Q$$

Profit Maximization

Profit maximization occurs when:

$$MR = MC$$

Managers use differentiation to find optimal output.

Q10 Write a brief note on marketing Research approaches to Demand estimation.

- **Introduction**

Demand estimation is one of the most important tasks in managerial economics. It helps firms determine:

- How much to produce
- What price to charge
- Future sales expectations
- Market opportunities

Demand estimation refers to predicting future demand for a product using various research methods.

Marketing research approaches help firms collect and analyze data about consumer behavior, market trends, and demand patterns.

- **Importance of Demand Estimation**

Demand estimation helps firms:

- Reduce business risk
- Plan production
- Set pricing strategies
- Make investment decisions
- Forecast future demand
- Improve profitability

Example:

If a company estimates high demand for smartphones, it will increase production.

- **Marketing Research Approaches to Demand Estimation**

There are several marketing research approaches used to estimate demand:

1. Consumer Survey Method

This method involves **collecting information directly from consumers** about their preferences and buying behavior.

Types of Consumer Surveys

(a) Complete Enumeration Survey

- Survey all consumers
- More accurate but costly
- Used when customers are limited

Example:

A company surveys all customers of a small town.

(b) Sample Survey

- Survey only a selected group
- Less costly
- Most commonly used method

Example:

Survey 500 consumers to estimate demand for a new product.

Advantages

- Direct information
- Accurate consumer preferences
- Useful for new products

Disadvantages

- Expensive
- Time-consuming
- Consumers may give inaccurate responses

2. Market Experiment Method

This method involves **testing products in a limited market** before launching nationwide.

Types of Market Experiments

(a) Controlled Experiment

Company changes variables like:

- Price
- Advertisement
- Packaging

Then measures demand.

Example:

Company tests two different prices in two cities.

(b) Test Marketing

Product is launched in a small market to observe demand.

Example:

A new beverage launched in Karachi before national launch.

Advantages

- Real market conditions
- Reliable data
- Helps reduce risk

Disadvantages

- Expensive
- Competitors may copy strategy

3. Statistical Method

Statistical methods use **historical data and mathematical models** to estimate demand.

Common Statistical Methods

(a) Regression Analysis

Used to find relationship between:

- Demand and price
- Demand and income
- Demand and advertisement

Example:

Demand = f (price, income, advertisement)

(b) Correlation Analysis

Measures relationship between variables.

Example:

Demand increases when income increases.

Advantages

- Scientific method
- Accurate predictions
- Useful for long-term forecasting

Disadvantages

- Requires skilled analysts
- Complex method

4. Delphi Method (Expert Opinion Method)

This method collects **opinions from experts** to estimate demand.

Experts include:

- Economists
- Market analysts
- Industry specialists

Procedure:

1. Experts give opinions
2. Responses collected
3. Final forecast prepared

Advantages

- Useful when data not available
- Good for long-term forecasting

Disadvantages

- Subjective
- Depends on expert judgment

5. End-use Method (Additional Method)

Demand is estimated based on **final use of product**.

Example:

Demand for steel depends on:

- Construction industry
- Automobile industry

6. Barometric Method

This method uses **economic indicators** to forecast demand.

Indicators include:

- GDP
- Income level
- Population growth
- Inflation

Example:

Increase in income → Increase in demand
